Journal of Clinical Immunology, Vol. 13, No. 4, 1993

Long-Term Use of IgA-Depleted Intravenous Immunoglobulin in Immunodeficient Subjects with Anti-IgA Antibodies

CHARLOTTE CUNNINGHAM-RUNDLES,[1,3] ZHUO ZHOU,[1] SAMIA MANKARIOUS,[2] and SUZANNE COURTER[2]

Accepted: March 16, 1993

The use of intravenous immunoglobulin is standard practice for antibody replacement in the humoral immunodeficiency diseases. Most infusions proceed uneventfully, but a proportion of infusions (5–8%) produces some degree of an infusion reaction. While the cause of most of these infusion reactions is unknown, an established, but rare cause of reactions is IgA antibodies in the serum of the patient, which apparently forms an immune complex with the traces of IgA in the infused immunoglobulin. This article describes our studies of five immunodeficient patients who had high-titered anti-IgA antibodies and a history of severe infusion reactions to intravenous immunoglobulin products not depleted of IgA (IgA content, 270–720 μg/ml). Over a 6-year period we gave these patients IgA-depleted intravenous immunoglobulin for a total of 170 infusions. These infusions were generally well tolerated; however, mild to moderate infusion reactions did occur in 9 of the 170 infusions (5.3%). These reactions were not related to the IgA content of the immunoglobulin solutions used—ascertained to vary between 0.4 and 2.9 μg/ml of IgA. Levels of plasma C3a and C4a increased after immunoglobulin infusions but the appearance of these components was not accompanied by any infusion reaction. We conclude that the long-term infusions of IgA-depleted intravenous immunoglobulin, within the range of IgA concentrations investigated, into patients with even very high-titered antibodies to IgA, is a safe practice.

KEY WORDS: IgA deficiency; hypogammaglobulinemia; anti-IgA antibodies; intravenous immunoglobulin; IgA content.

[1]Mount Sinai Medical Center, One Gustave Levy Place, New York, New York 10029.
[2]Baxter Healthcare Corporation, Hyland Division, Glendale, California 91203.
[3]To whom correspondence should be addressed at Box 1089, Mt. Sinai Medical Center, One Gustave-Levy Place, New York, New York 10029.

INTRODUCTION

Antibodies to IgA may be present in sera of as many as 40% of patients with selective IgA deficiency (1–6) or 10% of patients with common variable immunodeficiency (CVI) (4). Since the presence of this antibody can be associated with severe anaphylactic reactions to infusions of blood or blood products (1–6), the recognition of this antibody can be of clinical importance. Patients with CVI are often treated with intravenous immunoglobulin preparations which contain variable amounts of IgA, while most patients with selective IgA deficiency do not receive immunoglobulin prophylaxis, those who have IgG2 subclass deficiency and lack sufficient antibodies to bacterial carbohydrate antigens may benefit by treatment with intravenous immunoglobulin (7–9). In both cases, the therapeutic need for an IgA-depleted immunoglobulin product arises. We and others have reported that IgA-depleted intravenous immunoglobulin can be given in a few cases (10, 11) but the continued safety of this practice over time has not been ascertained. It is also not clear exactly how much depletion of IgA is necessary to avoid an infusion reaction in patients who have anti IgA antibodies.

In the past 12 years we have treated 105 patients with humoral immunodeficiency with intravenous immunoglobulin. In this period of time we have seen five patients who had a history of severe infusion reactions, who had high serum titers of anti-IgA antibodies. This article describes our evaluation and long-term medical management of such patients.

0271-9142/93/0700-0272$07.00/0 © 1993 Plenum Publishing Corporation

METHODS

Patients

In these studies five patients with either CVI or IgA deficiency and IgG subclass deficiency, who also had anti-IgA antibodies, were infused with an IgA-depleted intravenous immunoglobulin. Patient data are summarized in Table I. Patients 1, 2, 4, and 5 had a history of moderate to severe infusion reactions when previously given blood (one patient) and/or one to three infusions of various types of intravenous immunoglobulin solutions, not depleted of IgA. these infusion reactions are summarized in Table II. Reactions to these infusions included severe myalgias, shortness of breath and chest pain, fever, severe abdominal pain, and/or hypotension.

Anti-IgA Antibodies

Antibodies (of IgG isotype) to IgA were detected as previously described (10). Briefly, 96-well microtiter plates (NUNC, Vanguard International) were coated with 10 μg/ml of IgA myeloma LM or LAK (IgA1 or IgA2) in 0.1 *M* Na carbonate buffer. Serum from various patients or normal controls (10 to 15 samples in each assay), diluted 1:200 in 200 μl of PBS–0.2% Tween, was added to each well in duplicate, and after incubation for 3 to 5 hr at room temperature, the wells were washed, and goat anti-human IgG (1:2000 in PBS–Tween) (Sigma, St. Louis, MO) was added. (This reagent was chosen over three other commercial alkaline phosphatase anti-human IgG reagents tested since it had the lowest reactivity to purified IgA and IgM proteins.) After further incubation of microtiter plates overnight at room temperature, the washed wells were incubated at 37°C with nitrophenyl phosphate (Sigma), 1 mg/ml in 0.05 *M* Na carbonate-containing 0.1 m*M* $MgCl_2$. The absorbance at 405 nm was then read in a microtiter plate reader. Using this assay, sera were titered by serial twofold dilutions to determine the comparative amount of anti-IgA antibody in each sample. Sera from 20 normal controls, IgG from two pools of human immunoglobulin intended for intramuscular injection, and sera of 55 hypogammaglobulinemic subjects drawn prior to immunoglobulin infusion were tested in this assay; sera from all but the five patients discussed here were negative in this assay.

Immunoglobulin Infusions

The infused immunoglobulin was Gammagard (Baxter, Hyland Division, Glendale, CA). After the reactions experienced using non-IgA-depleted immunoglobulins and the finding of positive anti-IgA antibodies, patients were infused for 1 to 5 years with an IgA-depleted immunoglobulin at intervals of 3 to 4 weeks at a dose of 300 mg/kg body weight. A total of 28 lots of immunoglobulin was used with a measured level of IgA between 0.4 and 2.9 μg of IgA per ml,

Table I. Patients Treated with IgA-Depleted Intravenous Immunoglobulin

Patient No.	Age	Sex	Serum immunoglobulins (mg/dl) IgG	IgA	IgM	IgG subclass deficiency	Titer of anti-IgA antibody[a]	Clinical course	IgA-depleted IVGG, number infusions given	Immunoglobulin given (g/infusion)
1	54	F	54	<1	36	IgG1 to IgG4	1:7000	Recurrent respiratory infections	67	20
2	46	M	554	<1	28	IgG2	1:500	Recurrent bronchitis	47	20
3	13	M	449	26	58	IgG2 + IgG3	1:500	Recurrent respiratory infections, S/P Hodgkin's disease	30	12.5
4	27	F	210	<1	21	IgG1 to IgG4	1:200	Recurrent respiratory infections	8	20
5	54	F	270	7	90	IgG1 to IgG4	1:500	Recurrent sinusitis	20	20

[a]Titer of anti-IgA antibody was determined by ELISA as described (10). These titers were compared to sera of 50 other hypogammaglobulinemic subjects who had no detectable antibody to IgA, 20 normal control sera, and two pools of human immunoglobulin intended for intramuscular injection, suitably diluted. Titers were the same when using either the IgA1 and IgA2 myeloma proteins to coat ELISA plates.

Table II. History of Infusion Reactions

Patient No.	Immunoglobulin treatment[a]	Reactions
1	1. Intramuscular Ig, 1975–1990	None
	2. Sandoglobulin, 1980; 3 infusions	Severe abdominal pain, lumbar myalgias, chills, fever 101°C, shortness of breath, and chest pain on all 3 infusions, increasing symptoms with each infusion.
	3. Gamonative, 1983–1984; 16 infusions	None (reviewed in Ref. 10)
	4. Gammagard, 1985; 67 infusions, 3 reactions	As described in text
2	1. Blood transfusion, 1979	Hypotension, cyanosis, rigors, shortness of breath, abdominal pain, chest pain, sense of "impending doom"
	2. Sandoglobulin, 1986; 1 infusion	Hypotension, cyanosis, rigors, myalgias, sense of "impending doom"
	3. Gammagard; 1986–present; 47 infusions, 6 reactions	As described in text
4	1. Gamimmune-N, 1985–1986; 10 infusions	After 8 infusions, hypotension, myalgias, shortness of breath on the last 2 infusions
	2. Sandoglobulin, 1986; 1 infusion	Hypotension, myalgias, shortness of breath
	3. Gammagard, 1986–1987; 8 infusions	None
5	1. Sandoglobulin, 1988; 1 infusion	Severe myalgias, chills, fever 101°C
	2. Gammagard, 1988–present; 20 infusions	None

[a]Average IgA content in Sandoglobulin, 720 μg/ml (24); in Gamonative, 3–5 μg/ml (10); and in Gamimune-N, 270 μg/ml (24).

reconstituted as a 5% immunoglobulin solution. The number of grams of immunoglobulin given to each patient at each infusion is given in Table I. The solution was infused at 0.5 ml/min for 10 min, and the rate was increased to 1 ml/minutes for 1 hr, then increased to 1.5 to 25.0 ml/min as tolerated after this time.

Complement Activation Studies

Prior to studies of complement activation, all patients had been receiving the IgA-depleted immunoglobulin preparation for at least 5 months. Plasma samples were obtained from four patients prior to infusion of immunoglobulin and subsequently after specified time intervals. Blood samples were collected in Vacutainer tubes containing EDTA to prevent further complement activation. Anticoagulated blood samples were centrifuged at 4°C to remove cells and the plasma was frozen within 30 min of collection at −70°C. C3a and C4a were tested by an RIA using commercial kits for human complement C3a and C4a des Arg [^{125}I] (Amersham, Arlington Heights, IL). In principle, the test is based on the ability of a fixed amount of antibody to bind ^{125}I-labeled complement as an antigen, separating the bound from unbound antigen by precipitation, and centrifugation and counting the radioactivity of the labeled immune complex (12). A standard curve was prepared by holding the labeled antigen at a fixed concentration while increasing amounts of unlabeled antigen were added. Samples were quantitated by comparison to the standard.

In addition to the five patients who were infused with immunoglobulin and studied for complement activation at specified intervals, one additional hypogammaglobulinemic patient, who did not have an anti IgA-antibody, was infused with the same lot of immunoglobulin and tested as described above, as a control.

RESULTS

Anti-IgA Antibodies

The relative amounts of anti IgA-antibodies were compared by ELISA on serial, twofold diluted samples of serum on microtiter plates coated with the two myeloma IgA proteins, LM and LAK. The results for the various sera ranged from 1:200 to 1:7000 and were similar for each myeloma; these titers are given in Table I.

Infusion Reactions

During a 5-year period, 170 infusions with the IgA-depleted immunoglobulin were given to these five subjects. Most of these infusions were accompanied by no detectable infusion reactions; however, there were 9 significant reactions which occurred of the 170 infusions, (5.3%). Only two patients had reactions: patients 1 and 2, who had the highest titers of anti IgA-antibodies. None of these reactions were classified as severe; six reactions were moderate, with involuntary muscle contractions, shaking, chills, rigors, hives, or muscle aches, being the common symptoms. Three reactions were mild, with slight chills, and achiness being the manifestations. In each case, the infusions were stopped, and treatment with 100 mg of hydrocortisone and 50 mg diphenhydramine (moderate reactions) or diphenhydramine alone (mild reactions) was given.

To determine if the reactions were related to the amount of infused IgA, we retrospectively ascertained the IgA content of each lot of IgG and compared it to the number of reactions which occurred with each lot. From these comparisons we found that reactions took place with lots of immunoglobulin which contained various amounts of IgA, spanning a range from 0.4 to 1.1 μg/ml (Table III). Thus, there was no clear association between reactions and IgA content. Although there were a few lots which contained more than 2.0 μg/ml of IgA, by chance, Patients 1 and 2 did not receive these lots.

Table III. Lots Associated with Reactions

Content of IgA (μg/ml)	Lot No.	Number of reactions	Patient No.
0.4	25	1	1
0.6	21	2	2
0.7	20	1	2
0.8	13	4	1,2
1.1	11	1	2

Complement Activation

For first four patients studied, complement activation was investigated after one lot of immunoglobulin containing 0.4 μg/ml of IgA was administered. Calculating from the total dose of immunoglobulin infused, for patients 1, 2, and 4, the total amount of IgA infused was 160 μg; for patient 3, the amount of IgA infused was 100 μg. Prior to, during, and after this set of infusions, complement split products C3a and C4a were determined in plasma samples from each patient (Tables IV and V). Even prior to infusion, C3a and C4a were found increased in the plasma of all studied patients when compared to the amounts in plasma of normal donors. After infusion, all patients had evidence of complement activation as judged by increased C3a and C4a levels, with the greatest amounts of these products being found in the plasma of patient 4, who also had the largest amounts of C4a, even before the infusion was started. However, no clinical reaction of any kind during this set of infusions was observed for any patient.

For comparison, a fifth patient who did not have a detectable anti-IgA antibody was similarly infused

Table IV. C3a (ng/ml)[a] in Plasma of Patients Treated with Immunoglobulin

Time	Patient No. 1	2	3	4	Hypogammaglobulinemic patient with no anti-IgA antibody
Preinfusion	176	256	100	224	224
2.5 min	200	280	108	420	240
5.0 min	320	352	176	440	296
10.0 min	400	376	200	580	336
30.0 min	480	500	232	540	420
45.0 min	540	580	200	580	420
60.0 min	880	680	224	540	500
120.0 min	880	376	240	1240	480
180.0 min	376	420	192	540	280
240.0 min	320	280	136	440	208

[a]Normal range for C3a (ng/ml) in normal plasma is 150 ± 18 or 25–100 ng/ml. (18)

Table V. C4a Levels (ng/ml)[a] in Plasma of Patients Treated with Immunoglobulin

Time	Patient with Anti-IgA antibodies 1	2	3	4	Hypogammaglobulinemic patient with no anti-IgA antibody
Preinfusion	480	280	280	820	240
2.5 min	600	352	320	880	272
5.0 min	720	368	368	920	312
10.0 min	600	384	400	1000	304
30.0 min	780	420	420	1040	340
45.0 min	820	384	480	1120	328
60.0 min	880	384	520	1200	368
120.0 min	920	440	280	1120	600
180.0 min	720	440	340	1000	340
240.0 min	520	368	296	740	264

[a]Normal range for C4a in plasma is 8–150 ng/ml (19).

and tested for complement activation. These data are also included in Tables III and IV. Prior to, and after immunoglobulin infusion, elevated levels of C3a and C4a, in a range similar to that of other patients, was found.

DISCUSSION

In these analyses, five patients with anti-IgA antibodies had a history of severe infusion reactions when exposed to immunoglobulins not depleted of IgA. Aside from patient 1, the data for whom are published (10, 13), we cannot prove that the prior infusion reactions experienced by these patients were due to IgA-anti IgA interactions. However, the reactions were quite severe, high-titered antibodies to IgA were present in the serum of these patients (but not 50 other hypogammaglobulinemic patients who have had no such infusion reactions to IgA-containing products), and these five patients subsequently have tolerated IgA-depleted immunoglobulins. Thus, we believe that the most likely explanation is that IgA-anti IgA interactions were involved in the severe reactions observed.

However, whether or not the prior infusion reactions were caused by anti-IgA antibodies, the question of medical management is still an important one. Could these patients with anti-IgA antibodies tolerate months or years of treatment with an IgA-depleted immunoglobulin? Were the few infusion reactions we saw related to the infusion of more IgA on that day?

The successful infusion of IgA-depleted immunogloblin concentrates into immunodeficient patients who have anti-IgA antibodies and a history of severe infusion reactions has been previously reported by ourselves and others (6, 10, 11). From the available data, a few patients have been successfully treated even though small amounts of IgA are still being infused; the titer of anti-IgA antibody is not boosted after this treatment (6, 10). In the present studies, 170 infusions of an IgA-depleted intravenous immunoglobulin were given to 5 patients with anti-IgA antibodies over a period of 6 years. The overall reaction rate for these infusions (5.3%) was similar to that often quoted for hypogammaglobulinemic patients as a whole (14–17). This number of infusion reactions was also not different overall from the incidence of reactions encountered by our large group of 105 hypogammaglobulinemic patients who do not have anti-IgA antibodies. Thus the infusions in this group did not produce more reactions than the number generally expected. However, hypogammaglobulinemic patients tend to have reactions at the start of intravenous immunoglobulin therapy (14, 15) but rarely afterward. The patients treated here had reactions intermittently during the months or years in which they have been treated. Because we knew that the immunoglobulin solutions contained varying amounts of IgA, our initial impression was that reactions were more likely to occur if the lot of immunoglobulin used on that date contained more IgA.

To determine if this was true or not, we retrospectively ascertained the IgA content of each of the lots of intravenous immunoglobulin used and compared this to the clinical observations made during and after each infusion. Only two patients had reactions; those patients who had the highest titers of anti-IgA antibodies. Patient 1 received six lots of immunoglobulin which contained 1.1 μg/ml IgA and patient 2 received five lots of similar concentration; neither received any lots containing more IgA than this amount. Only patient 2 had a reaction associated with one infusion of this group, while patient 1 had no reaction to these six infusions. Patients 1 and 2 had a total of four reactions to lots containing 0.8 μg/ml and scattered reactions to infusions containing less than this amount. Thus we concluded that the actual IgA content, at least in this range, was not a determining factor for an infusion reaction.

The presence of an anti-IgA titer of 1:500 in the serum of patient 3 was somewhat unexpected since this patient had an IgA serum level of 26 mg/dl. While we assume that it is directed to specific antigenic epitopes of IgA not possessed by the patient, we did not find that this antibody was directed exclusively to either IgA1 or IgA2. Another unusual situation is presented by patient 2, who tolerated intramuscular injections of immunoglobulin for 10 years with no reactions but who had severe reactions to an intravenous immunoglobulin not depleted of IgA (Sandoglobulin). We assume that these reactions were due to the larger amounts of IgA administered in the infused immunoglobulin.

The cause of infusion reactions in general is unfortunately still poorly understood. Certainly a major factor determining the likelihood of infusion reactions is the rate of infusion (14–17). The reactions experienced by these patients could have conceivably been due to infusion speed. While the infusion of aggregates may have produced some of the adverse effects reported by Barandun *et al.* (14), present IVGG preparations are almost devoid of such aggregates.

We have previously found that many hypogammaglobulinemic patients who are treated with IVGG do have immune complexes formed *in vivo* (18). Although all of the antigens are not elucidated, dietary sources provide part of the antigens involved (18). After patients are infused with immunoglobulin (as were the patients studied here), the amount of immune complex tends to diminish progressively (18). However, we have not found that the occurrence of infusion reactions is connected with immune complex formation since many infusions, for which immune complexes are found, proceed with no evidence of reactions (18). While we cannot specify the cause of the reactions seen in these studies, one potential cause, infusion of higher levels of IgA on these dates, can probably be eliminated.

Ferreira *et al.* (6) have also infused patients with known IgA antibodies with IgA-depleted intravenous immunoglobulin; however, the amounts of IgA in these infusions were greater than those in the solutions we used. The lots tested in that study contained between 20 and 70 μg/ml, more than 20 times the IgA level in most of our infusions. Seven patients of the treated group in the Ferreira study had anti-IgA antibodies; all but two had very low titers. The success of the infusions in six of these patients, despite the amount of infused IgA, could have been due to the low titer of anti-IgA antibody in these subjects.

In our analyses, we also studied if complement activation could be documented by measurement of C3a and C4a before, during, and after a selected group of immunoglobulin infusions. While normal amounts of C3a and C4a in human serum have not been definitively settled, normal values for C3a determined by radioimmuno assays similar to the one used here have been 150 ± 70 ng/ml (19) or 25–100 ng/ml (20); for C4a a level of 8–150 ng/ml has been cited (19). For the hypogammaglobulinemic patients with anti-IgA antibodies studied here, levels of C3a and C4a were elevated even before infusions were started and increased afterward for each patient up to a maximum of 1240 ng/ml for C3a and 1200 ng/ml for C4a for one patient. While these levels are certainly increased, C3a measured in asymptomatic patients receiving hemodialysis treatment was 2907 ± 372 ng/ml, whereas patients experiencing recurrent allergic reactions can have much higher levels of C3a, 8533 ± 157 ng/ml (21). The absence of clinical reactions in our patients was thus not surprising since the C3a levels observed were well below the expected symptomatic range. Additionally, one hypogammaglobulinemic patient who did not have a detectable anti-IgA antibody was tested for complement split products. After intravenous infusion, small increases over preinfusion levels were found at 60 and 120 min after the start of the infusion. The complement activation seen in this patient was similar to that found for patients known to have anti-IgA antibodies. The cause of this mild complement activation in these patients is obviously unknown, but for patients with anti-IgA antibodies, the formation of IgG–IgA complexes would be a possible reason. Such complexes have been observed previously to appear in this clinical situation (13, 22).

While there are many unresolved issues concerning anti-IgA antibodies, we have concluded that most patients with even high-titered anti-IgA antibodies can be safely infused with IgA-depleted intravenous immunoglobulin solutions for prolonged periods. While the rare development of anti-IgA antibodies of the IgE class can develop in these circumstances which may lead to reactions (6, 23), such patients can also be successfully infused if the IgA content of the infused solution is sufficiently reduced (23). In our studies, patients with very high titers of anti-IgA antibodies of the IgG class had sporadic mild to moderate reactions, but the total amount of IgA infused, within the range 0.2 to 1.1 μg IgA/ml, was apparently not the causative factor.

REFERENCES

1. Vyas GN, Perkins HA, Fudenberg HH: Anaphylactoid transfusion reaction associated with anti-IgA. Lancet 2:312–314, 1971
2. Schmidt AP, Taswell HF, Gleich GJ: Anaphylactic transfusion reactions associated with anti-IgA antibody. N Engl J Med 280:188–193, 1969
3. Rivat I, Rivat C, Daveau M, Ropartz C: Comparative frequencies of anti-IgA antibodies among patients with anaphylactic transfusion reactions and among normal blood donors. Clin Immunol Immunopathol 7:340–348, 1977
4. Wells JV, Buckley RH, Schanfield MS: Anaphylactoid reactions to plasma infusions in patients with hypogammaglobulinemia and anti-IgA antibodies. Clin Immunol Immunopathol 8:265–271, 1971
5. Hammarström L, Persson MAA, Smith CIE: Anti-IgA in selective IgA deficiency *in vitro* effects and Ig subclass patterns of human anti-IgA. Scand J Immunol 18:509–513, 1983
6. Ferreira A, Garcia-Rodriguez MC, Lopez-Trascasa M, Pascual-Salcedo D, Fontan G: Anti-IgA antibodies in selective IgA deficiency and in primary immunodeficient patients treated with gamma-globulin. Clin Immunol Immunopathol 47:199–207, 1988
7. Oxelius VA, Laurell AB, Lindqvist B, Golebiowska M, Axelsson U, Björkander J, Hanson LÅ: IgG subclasses in selective IgA deficiency. Importance of IgG2-IgA deficiency. N Engl J Med 302:1476–1477, 1981
8. Oxelius VA: Immunoglobulin (IgG) subclasses and human disease. Am J Med 76:7–18, 1984
9. Björkander J, Hammarström L, Smith CI, Buckley RH, Cunningham-Rundles C, Hanson LÅ: Immunoglobulin prophylaxis in patients with antibody deficiency syndromes and anti-IgA antibodies. J Clin Immunol 7:8–15, 1987
10. Cunningham-Rundles C, Wong S, Björkander J, Hanson LÅ: Use of an IgA-depleted intravenous immunoglobulin in a patient with an IgA antibody. Clin Immunol Immunopathol 38:141–9, 1986
11. Björkander J, Bengtsson U, Oxelius VA, Hanson LÅ: Symptoms in patients with lowered levels of IgG subclasses, with or without IgA deficiency and effects of immunoglobulin prophylaxis. Monogr Allergy 20:157–163, 1986
12. Hugli TE, Chenoweth DF: Biologically active peptides of complement: Techniques and significance of C3a and C5a measurements. *In* Laboratory and Research Methods in Biology and Medicine, RM Nakamusa, WR Dito (eds.). New York, Alan R. Liss, 1980, p 443
13. Day NK, Good RA, Wahn V: Adverse reaction in selected patients following intravenous infusions of gammaglobulin. Am J Med Suppl 3A:25–31, 1984
14. Barandun S, Morrell A: Adverse reactions to immunoglobulin preparations. *In* Immunohemotherapy, UE Nydegger (ed). London, Academic Press, 1981, pp 223–230
15. Cunningham-Rundles C, Siegal FP, Smithwick EM, Cunningham-Rundles S, Lion-Boule A, O'Malley J, Barandun S, Good RA: Efficacy of intravenous gammaglobulin in humoral immunodeficiency disease. Ann Intern Med 101:435–439, 1984
16. Ammann AJ, Ashman RF, Buckley RH, Hardie WR, Krantman HJ, Nelson J, Ochs H, Stiehm ER, Tiller T, Wara DW, Wedgewood R: Use of intravenous gammaglobulin in antibody immunodeficiency: Results of a multicenter controlled trial. Clin Immunol Immunopathol 22:60–67, 1982
17. Stiehm ER, Ashida E, Kim KS, Winston DJ, Haas A, Gale RP: Intravenous immunoglobulin as therapeutic agents. Ann Intern Med 107:367–382, 1987
18. Cunningham-Rundles C, Carr RI: Dietary bovine antigens and immune complex formation after intravenous immunoglobulin in common variable immunodeficiency. J Clin Immunol 6:381–388, 1986
19. Zilow G, Sturm JA, Rother U, Kirschfink M: Complement activation and the prognostic value of C3a in patients at risk of adult respiratory distress syndrome. Clin Exp Immunol 79:151–157, 1990
20. Sundsmo JS, Papin RA, Wood L, Hirani S, Waldeck N, Buckingham B, Kershnar A, Ascher M, Charles MA: Complement activation in Type 1 human diabetes. Clin Immunol Immunopathol 35:211–225, 1985
21. Hakim RM, Breillatt J, Lazarus MJ, Port F: Complement activation and hyper-sensitivity reactions to dialysis membranes. N Engl J Med 311:878–882, 1984
22. Cunningham-Rundles C, Day NK, Wahn V, Smithwick EM, Siegal FP, Gupta S, Good RA: In Reaction to Intravenous Gammaglobulin Infusions and Immune Complex Formation, Immunohemotherapy, UE Nydegger (ed). London, Academic Press, London, 1981, pp 447–449
23. Burks AW, Sampson HA, Buckley RH: Anaphylactic reactions after gammaglobulin administration in patients with hypogammaglobulinemia. N Engl J Med 314:560–562, 1986
24. Romer J, Morgenthaler JJ, Scherz R, Skvaril F: Characterization of various immunoglobulin preparations for intravenous application. I. Protein composition and antibody content. Vox Sang 42:62–73, 1982